

KRYPTOS

(ciphertext)

EMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ
YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD
VFPJUDEEHZWETZYVGWHKKQETGFQJNCE
GGWHKK?DQMC PFQZDQMMIAGPFXHQR LG
TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA
QZGZLECGYUXUEENJTB JLBQCRTBJDFHRR
YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI
HHDDDUVH?DWKB FUF PWNTDFIYCUQZERE
EVLDKFEZMQOQJLTTUGSYQPF EUNLAIDX
FLGGTEZ?FKZBSFDQVGOGI PUFXHHDRKF
FHQNTGPUAECNUVPDJMQCLQUMUNEDFQ
ELZZVRRGKFFVOEEXBDMV PNFQXEZLGRE
DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP
DQUMEBEDMHDAFMJGZNUPLGESWJLLAETG

(Vigenere tableau)

ABCDEFGHIJKLMNOPQRSTUVWXYZABCD
 AKRYPTOSABCDEFGHIJLMNQUVWXZKRY P
 BRYPTOSABCDEFGHIJLMNQUVWXZKRYPT
 CYPTOSABCDEFGHIJLMNQUVWXZKRYPT O
 DPTOSABCDEFGHIJLMNQUVWXZKRYPTOS
 ETOSABCDEFGHIJLMNQUVWXZKRYPTOSA
 FOSABCDEFGHIJLMNQUVWXZKRYPTOSAB
 GSABCDEFGHIJLMNQUVWXZKRYPTOSABC
 HABCDEFGHIJLMNQUVWXZKRYPTOSABCD
 IBCDEFHGIJLMNQUVWXZKRYPTOSABCDE
 JCDEFHGIJLMNQUVWXZKRYPTOSABCDEF
 KDEFHGIJLMNQUVWXZKRYPTOSABCDEFG
 LEFGHGIJLMNQUVWXZKRYPTOSABCDEFGH
 MFGHGIJLMNQUVWXZKRYPTOSABCDEFGHI

ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA
CHTNREYULDSLSSLNHOHSNOSMRWXMNE
TPRNGATIHNRRAPESLNNELEBLPIIACAE
WMTWNDITEENRAHCTENEUDRETNHAE
TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR
EIFTBRS PAMHHEWENATAMATEGYEERLB
TEEFOASF IOTUETUAEOTOARMAEERTNRTI
BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB
AECTDDHILCEIHSITEGOEAOSSDDRYDLORIT
RKLMLIEHAGTDHARDPNEOHTGMFMEUHE
ECDMRIPEFMEIHNLSSTHRTVDOHW?OBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

NGHIJLMNQUVWXZKRYPTOSABCDEFGHIJL
OHIJLMNQUVWXZKRYPTOSABCDEFGHIJL
PIJLMNQUVWXZKRYPTOSABCDEFGHIJLM
QJLMNQUVWXZKRYPTOSABCDEFGHIJLMN
RLMNQUVWXZKRYPTOSABCDEFGHIJLMNQ
SMNQUVWXZKRYPTOSABCDEFGHIJLMNQU
TNQUVWXZKRYPTOSABCDEFGHIJLMNQUV
UQUVWXZKRYPTOSABCDEFGHIJLMNQUVW
VUVWXZKRYPTOSABCDEFGHIJLMNQUVWX
VWXZKRYPTOSABCDEFGHIJLMNQUVWXZ
XWXZKRYPTOSABCDEFGHIJLMNQUVWXZK
YZKRYPTOSABCDEFGHIJLMNQUVWXZKR
ZZKRYPTOSABCDEFGHIJLMNQUVWXZKRY
ABCDEFIGHIJKLMNOPQRSTUVWXYZABCD

Defects:

S = unintentionally left out of the sculpture

L = unintentionally added to the sculpture

DY_{AHR} = irregularly positioned

K0

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EEVIRTUALLYE      . . . . - . . . . . - . . . - . - . .
EEEEEEINVISIBLE   . . . . . . . - . . . . - . . . . . - . . . .

    EESHADOWEE     . . . . . . . - . - . . - . - . . . .
FORCESEEEEE       . . - . - . . - . - . . . . . . . . .

    LUCIDEE        . - . . . - . - . . . - . . . . .
    MEMORYE        - . . - . - . - . . - . - . . .

            RQ              . - . - . - .
           SOS             . . . - . . .

EDIGETALEEEE      . - . . . - . . - . - . - . . . .
INTERPRETATIT     . . - . - . . - . - . - . . - . . -

    TISYOUR        - . . . . - . - . - . . . . .
    POSITIONE       . - . . - . . . . - . . - . - . .

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Why so many E's? Is this just background noise in the transmission? What happens if we filter out the noise? 26xE's are superfluous and 6xE's are intelligible parts of the words with the exception of the misspelled word DIGETAL. The filter doesn't distinguish which E's are important...

VIRTUALLY, INVISIBL, SHADOW, FORCS, LUCID, MMORY, RQ, SOS, DIGTAL, INTRPTATIT, TISYOUR, POSITION

SOS is a palindrome. (note: palindrome shares a dictionary page with the word PALIMPSEST). A palindrome is a string of letters that are the same when read forwards or backwards. It could be that the "Central Intelligence Agency" is itself a clue about this puzzle which is physically located at the absolute center of the property. Looking at the central letters of each palindrome element found in the morse gives,

LL, ISI, MM, SOS, RPR, TAT, TIT, ITI

The central letter results: L S M O P A I T (note: had the word DIGETAL been spelled correctly, it would've contributed a G by the palindrome fragment IGI from DIGITAL.)

K1

EMUFPHZLRFAXYUSDJKZLDRNSHGNFIVJ
YQTQUXQBQVYUVLLTREVJYQTMKYRDMFD

K1 is solved using the tableau and keyword PALIMPSEST in what is known as a Quagmire III cipher:

BETWEENSUBTLESHADINGANDTHEABSENCEOFFLIGHTLIESTHENUANCEOFFILUSION
1 23 4 5 6 7 8 9 10

This message is 63 long. I think its goal is to communicate something important by looking at the relative middle point with respect to the misspelled letter. The Q is at the 57th position. The center of 1-56 is 28, which is also where ABS is positioned. This 3-letter group is also indicated by being grouped in the center between the 5th and 6th E's in the string. Does Jim Sanborn then use a dictionary to find a keyword by looking up ABS?

K2

VFPJUDEEHZWETZYVGWHKKQETGFQJNCE
GGWHKK?DQMCFFQZDQMMIAGPFXHQRLG
TIMVMZJANQLVKQEDAGDVFRPJUNGEUNA
QZGZLECGYUXUEENJTBQLBQCRTBJDFHRR
YIZETKZEMVDUFKSJHKFWHKUWQLSZFTI
HHDDDUVH?DWKBFUFPWNTDFIYCUQZERE
EVLDKFEZMOQQJLTUGSYQPFEUNLAVIDX
FLGGTEZ?FKZBSFDQVGOGIPUFXXHDKRF
FHQNTGPAECNUVPDJMQCLQUMUNEDFQ
ELZZVRRGKFFVOEEXBDMVPNFQXEZLGRE
DNQFMPNZGLFLPMRJQYALMGNUVPDXVKP
DQUMEBEDMHDAFMJGZNUPLGESWJLLAETG

K2 is also solved by Quagmire III, this time using the keyword ABSCISSA:

IT WAS TOTALLY INVISIBLE HOWS THAT POSSIBLE? THEY USED THE EARTHS MAGNETIC FIELD X THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGROUND TO AN UNKNOWN LOCATION X DOES LANGLEY KNOW ABOUT THIS? THEY SHOULD ITS BURIED OUT THERE SOMEWHERE X WHO KNOWS THE EXACT LOCATION? ONLY WW THIS WAS HIS LAST MESSAGE X THIRTY EIGHT DEGREES FIFTY SEVEN MINUTES SIX POINT FIVE SECONDS NORTH SEVENTY SEVEN DEGREES EIGHT MINUTES FORTY FOUR SECONDS WEST X LAYER TWO

Could it be that K2 is similar to K1 in its use of a misspelling in order to point to specific position in the text? The U is at position 116. The center of 1-115 is 57.5. There is an A at 57 and a G at 58. The A and G are found in the center group of letters following the 6th E out of 13 E's,: ARTHSMAGN. The letters AG might be a special clue to a keyword for K4. The three nearest words of interest in the dictionary when looking up AG are: (1)Ag, "Argentum", the latin word for silver, (2) agapé, an ancient Greek word for divine or preferred love, and (3) agency, which is part of the acronym, C.I.A.

K3

ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA
CHTNREYULDSLSSLNOHSNOSMRWXMNE
TPRNGATIHNRRAPESLNNELEBLPIIACAE
WMTWNDITEENRAHCTENEUDRETNHAE
TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR
EIFTBRSPAMHHEWENATAMATEGYEERLB
TEEFOASFOTUETUAEOTOARMAEERTNRTI
BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB
AECTDDHILCEIHSITEGOEASDDRYDLORIT
RKLMLEHAGTDHARDPNEOHMGFMFEUHE
ECDMRIPFEIMEHNLSSTTRTVDOHW

K3 is solved by applying transpositions first by mapping the code to a 24x14 matrix, turning 90-degrees counterclockwise and then fitting it to an 8x42 matrix and turn.

ENDYAHROHNLSRHEOCPTEOIBIDYSHNAIA
CHTNREYULDSLSSLNOHSNOSMRWXMNE
TPRNGATIHNRRAPESLNNELEBLPIIACAE
WMTWNDITEENRAHCTENEUDRETNHAE
TFOLSEDTIWENHAEIOYTEYQHEENCTAYCR
EIFTBRSPAMHHEWENATAMATEGYEERLB
TEEFOASFOTUETUAEOTOARMAEERTNRTI
BSEDDNIAAHTTMSTEWPIEROAGRIEWFEB
AECTDDHILCEIHSITEGOEASDDRYDLORIT
RKLMLEHAGTDHARDPNEOHMGFMFEUHE
ECDMRIPFEIMEHNLSSTTRTVDOHW?OBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

NGHIJLMNQUVWXZKRYPTOSABCDEFGHIJ
OHIJLMNQUVWXZKRYPTOSABCDEFGHIJL
PIJLMNQUVWXZKRYPTOSABCDEFGHIJLM
QJLMNQUVWXZKRYPTOSABCDEFGHIJLMN
RLMNQUVWXZKRYPTOSABCDEFGHIJLMNQ
SMNQUVWXZKRYPTOSABCDEFGHIJLMNQ
TNQUVWXZKRYPTOSABCDEFGHIJLMNQ
UQUVWXZKRYPTOSABCDEFGHIJLMNQ
VUVWXZKRYPTOSABCDEFGHIJLMNQ
WVWXZKRYPTOSABCDEFGHIJLMNQ
XWXZKRYPTOSABCDEFGHIJLMNQ
YXZKRYPTOSABCDEFGHIJLMNQ
ZZKRYPTOSABCDEFGHIJLMNQ
ABCDEFGHIJKLMNQRSTUUVWXYZABCD

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q

This is borrowed from a Nov. 26th, 1922 journal of Howard Carter's discovery of King Tut's tomb. Minor alterations seem to indicate select line lengths were sought, thus the locations or spelling of certain letters or words was necessary for some purpose. I have some analysis on the passage that shows some evidence that the E's were tailored:

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT
 DESPERATELY
 ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH
 WERE
 TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND
 CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE
 21
 CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER
 CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM
 CAUSING
 WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q
 42 44
 45 47

I'm looking at two or three possibilities: (1) are the E's tailored in such a way that they perform the same function as they do in K1 and K2, or (2) does the message hide a key number in the sum of all E's known from the original text, which would make it 47, or (3) is it doing both? This must be confusing, but I'm also making an assumption to ignore the two E's in the last portion for the purpose of identifying the central HOL clue, which may not really be a clue at all if this turns out to be just a simple count of E's by recognizing the subtle differences between what's on the sculpture and the original record of Howard Carter's journal. I hesitate to jump to conclusions like this, but the number 47 just happens to be the atomic number for Silver, Ag. This is both surprising and confusing. I still don't know why the X and Q are present in the passage.

K4

?OBKR
 UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
 TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
 VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

This is what's left to be deciphered. A ?-mark and 97 letters encrypted by some unknown process. Jim Sanborn and his cryptographic advisor Ed Scheidt have revealed something about their unique process: (1) it's masked, meaning something has been done to alter the English language, thwarting all standard methods of cryptanalysis, (2) five or six steps are used in the overall process of decryption, (3) letters 64-74: NYPVTTMZFPK are 1:1 decrypted to BERLINCLOCK. It is the hope that the crib tip BERLINCLOCK will speed up attempts to solve the remaining K4 ciphertext.

The Mask

One of the most perfect methods for masking a ciphertext is to use a source of random letters, one random letter for each letter in the cipher. This process is known as a One Time Pad (OTP). In terms

of frequency and distribution, any text can be analyzed for known consistencies for the language used. It's practically impossible to perceive any consistencies when one message has been paired with an unknown source. If the purpose of the work preceeding K4 is to provide the needed intelligence to solve K4, then it is my belief that the mask is recovered from somewhere in that data. It could be a specific range of values in order, or an algorithm of some disrupted pattern or something like a grille.

Here are two interesting ideas I'd like to consider: (1) making use of the odd letters resulting from the palindrome fragments of K0, and (2) applying the Period Table of the Elements.

From K0 we get palindrome letters:

L S M O P A I T anagrams to ALMOST PI

I've left to an interesting conclusion which follows just enough reasoning to look consistent with the entirety of K1 and K2 when considered together. Pi is that relationship between a circle's circumference and it's diameter. "Almost Pi" could be the common approximation for Pi, simply produced by 22 divided by 7, which is 3.14285714857142857... I discovered that this number sequence can be applied to K1/K2 to produce exactly 98 random looking letters.

$3 = 3^{\text{rd}}$ letter
 $3+1 = 4^{\text{th}}$ letter
 $3+1+4 = 8^{\text{th}}$ letter
 $3+1+4+2 = 10^{\text{th}}$ letter
 $3+1+4+2+8 = 18^{\text{th}}$ letter
 $3+1+4+2+8+5 = 23^{\text{rd}}$ letter
 $3+1+4+2+8+5+7 = 30^{\text{th}}$ letter
 $3+1+4+2+8+5+7+1 = 31^{\text{st}}$ letter...

(98 places, note: the rainbow convention is just a convenience)

I don't know which is preferred, using the plaintext or the ciphertext, so I'll try both, forwards and backwards given the speculation that the ?-position may suggest a reversal,

BETWEEN SUBTLE SHADING AND THE ABS
NCE OF FLIGHT LIES THE NUANCE OF I
LUSION IT WAS TOTALLY INVISIBL
HOWS THAT POSSIBLE? THEY USED TH
E EARTH'S MAGNETIC FIELD X THE IN
FORMATION WAS GATHERED AND TRAN
MITTED UNDERGROUND TO AN UNKNOW
N LOCATION X DOES LANGLEY KNOW AB
OUT THIS? THEY SHOULD ITS BURIED
OUT THERE SOMEWHERE X WHO KNOWS T
HE EXACT LOCATION? ONLY WW THIS
AS HIS LAST MESSAGE X THIRTY EIGH
T DEGREE FIFTY SEVEN MINUTES S

XPOINT FIVE SECONDS NORTH SEVEN
 TY SEVEN DEGREES EIGHT MINUTES
 FORTY FOUR SECONDS WEST XLAYER TW
 O

Forwards from PT:

TWSBIDENFITAQLOIANEHTALEHETSCDFOTOHASMEUUAWNAILEB
 OHSOTDOHREH THATNYWASAGIHTREEI IXNFNRNTVNEMFOFUSXWO

Backwards from PT:

OWXSUFOFMENV TNRN FNXI IEERTHIGASAWYNTAHTHERHODTOSHO
 BELIANWAUUEMSAHOTOFDCSTEHALETHENA IOLQATIFNEDIBSWT

Pulling mask out of the K1/K2 ciphertext(CT),

EMUFPHZLRFAXYUSDJ KZLDKRN SHGNFI
 VJYQTQUXQBQVYU VLLTREVJYQTMK
 YRDMFDVFPJUDEEHZ WETZYVGWHKK
 QETGFQJNCEGGWHKK? DQMC PFQZDQ
 MMIAGPFXHQRLGTIMVMZJANQLVKQ
 EDAGDVERPJUNGEUNAQQZGZLECGYU
 XUEENJTB JLBQCR TBJDFHRRY IZET
 KZEMVDUFKSJHKFWHKUWQLSZFTIH
 HDDDUVH?DWKBFUFPWNTDFIYCUQZ
 EREEVLDKFEZMOQQJLTTUGSYQPFE
 UNLAVIDXFLGGTEZ?FKZBSFDQVGO
 GIPUFXXH HDRKFFHQNTG PUAECNUVP
 DJMQCLQUMUNEDFQELZZVRRGKFFV
 OEEXBDMVPNFQXEZLGREDNQFMPNZ
 GLFLPMRJQYALMGNUVPDXVKPDQUM
 EBEDMHD AFMJGZNUPLGESWJLLAET
 G

Forwards from CT:

UFLFKRIVTUVKEYFVHZKQFJKMQMGFIJQEDFUGUXNTTHTKVUWQH
 HUHFDZEVDQUEUVDZBOGFHQUPDCQQVVOBMZDZGPRNXMEMDUSTG

Backwards from CT:

GTSUDMEMXNRPGZDZMBOVVQQCDPUQHFGOBZDVUEUQDVEZDFHUH
 HQWUVKTHTTNXUGUFDEQJIFGMQMKJFQKZHV FYKEVUTVIRKFLFU

Using the Period Table of the Elements,

A standard periodic table of elements, color-coded by groups. The title 'Periodic Table of the Elements' is centered at the top. The table includes all elements from Hydrogen (H) to Oganesson (Og), with Lanthanide and Actinide series shown separately at the bottom.

The following chain of letters results beginning with Hydrogen and ending with Cesium:

HHELIBECNOFNENAMGALSIPSC LARKCASCTIVCRMNFECONICUZ
 NGAGEASSEBRKRBSRYZRNBMOTCRURHPDAGCDINSNSBTEIXECS

This is purely a guess. I like that it results in the final letters coming from Cesium which is known to be the most accurate source for time keeping. Also, my work with tracking the E's in the deciphered section of K2 resulted in some speculation about why it produced the letters AG, possibly a hint to look at the abbreviations in a chart of the elements given that Ag represents the element Silver.

Mask Removal

This task might be more tricky than we initially think, since it's typical to assume one string of letters is being used to modify another string, letter for letter. The trick that may be involved may involve a spatial component. This could be done in a number of ways:

Typical lining up of rows:

A B C D (mask)
 &
a b c d (plaintext)
 Aa Bb Cc Dd (K4)

Horizontally reflected:

A B C D
 E F G H
 &
 a b c d
e f g h
 Ae Bf Cg Dh
 Ea Fb Gc Hd

Rotated and layered:

Ah Bg Cf De

Ed Fc Gb Ha

In order to consider these spatial possibilities we look at the factors of 98. 49x2, 14x7, 7x14 and 2x49 will return different results when reflecting or rotating,

49x2:

?OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJQSSEKZZ
WATJKLUDIAWINFBNYPVTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

14x7:

?OBKRUOXOGHULB
SOLIFBBWFLRVQQ
PRNGKSSOTWTQSJ
QSSEKZZWATJKLU
DIAWINFBNYPVTT
MZFPKWGDKZXTJC
DIGKUHUAUEKCAR